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Technology Insertion Is a Management System, Not a Demo Day

Defense organizations do not have a shortage of demonstrations.

They have a shortage of transition systems.

Every few months, a promising technology appears in front of a mission organization. The demo works. The model is interesting. The dashboard is clean. The autonomy behavior is impressive. The vendor can explain the architecture. The operators can imagine a use case. The executives can imagine a press release.

Then nothing scales.

The failure is usually not that the technology had no value. The failure is that technology insertion was treated as an event instead of a management system.

A demo can create belief. It cannot, by itself, create adoption.

The Demo Is Only One Evidence Point.

A demonstration answers a narrow question: can this capability do something useful under arranged conditions?

That matters. It is not the transition decision.

A transition decision has to connect mission ownership, workflow change, system integration, data availability, cyber posture, test evidence, sustainment, funding, and contracting. If those pieces are missing, the demo becomes a disconnected proof point. It may be true. It may even be impressive. It still has nowhere to go.

A familiar failure pattern looks like this: a team runs a successful pilot for an AI-enabled planning, sensing, logistics, or decision-support tool. Operators like it. Leaders like the briefing. The vendor leaves with a strong quote and a polished deck. Then the effort stalls because no one owns the operational workflow, the tool has not been mapped to the system boundary it must integrate with, the acceptance criteria were invented after the demonstration, and the only funding path was the pilot itself.

That is demo-day logic.

The technology worked. The management system did not.

Mission-Problem Ownership Comes First.

Technology insertion has to start with a named mission problem and a real owner.

“We need AI” is not a mission problem. “We need a better common operating picture” is usually not specific enough either.

A real problem has an operational context, a decision, a constraint, and a consequence. It names who is making the decision, what information is missing, what delay or risk is accumulating, what workaround exists today, and who has the authority to change the workflow.

Without that owner, pilots become orphaned. Everyone can like the idea. Nobody has to adopt it.

Mission ownership also changes the shape of the evaluation. The question becomes less “Is this technology interesting?” and more “Does this capability change a decision that matters enough to justify integration, training, cyber review, sustainment, and funding?”

That is the first gate.

Fit Beats Novelty.

Novel technology gets attention. Fit creates adoption.

Fit has three parts.

Technical fit asks whether the capability can live inside the target architecture, use the available data, satisfy identity and cybersecurity constraints, produce observable behavior, and survive the actual deployment environment.

Operational fit asks whether the capability changes a real workflow without breaking tempo, trust, human judgment, training, or degraded operations.

Acquisition fit asks whether there is a contract path, funding path, evaluation strategy, data-rights posture, and transition owner that can carry the effort after the pilot.

Most failed insertions do not fail only on technical performance. They fail at the boundary between those three forms of fit. A prototype may work on clean data but not on operational data. It may solve a real problem but require an authority change no one is prepared to make. It may be mature enough to test but trapped in a funding path that cannot buy the next phase.

The DoD Software Acquisition Pathway exists because software-heavy capability needs iterative delivery, user engagement, cybersecurity, and continuous improvement built into the acquisition model [1]. A software-heavy insertion effort that ignores those realities is already misdesigned.

Digital engineering points in the same direction. DoD digital engineering policy emphasizes authoritative sources of truth, models, data, and engineering information that inform lifecycle decisions [2]. Technology insertion should use the same discipline. The insertion case should be model-informed, evidence-backed, and connected to the program decision system.

Fit is not a vibe. It is a design constraint.

Acceptance Criteria Must Be Designed Before Enthusiasm Peaks.

Many pilots fail because nobody defines success until after the pilot is politically inconvenient.

By then, every answer is contaminated. The sponsor wants the effort to look promising. The vendor wants a continuation path. Operators may want the tool but not the integration burden. Executives may want the story more than the adoption work.

Acceptance criteria have to be set before the demonstration becomes a transition decision.

Good criteria name the operational delay or risk being reduced. They identify the system boundary, data condition, user workflow, cyber posture, test authority, sustainment owner, and next funding or contract path. They also name the stop condition.

That last part matters.

Without stop criteria, every pilot becomes a negotiation. With stop criteria, the organization can learn without pretending.

The GAO Technology Readiness Assessment Guide is useful because it treats maturity as evidence

demonstrated against a specific context, not enthusiasm asserted in the abstract [3]. Technology insertion needs the same posture. A demo does not prove maturity unless the environment, constraints, users, and evidence match the transition problem.

Transition Ownership Is a Design Requirement.

Every serious insertion effort needs a transition owner.

Not a sponsor. Not an interested observer. Not a senior executive who likes the idea.

A transition owner is accountable for carrying the capability from pilot to adoption. That owner has to connect the mission organization, technical environment, acquisition path, funding plan, cybersecurity posture, test evidence, sustainment model, and user community.

If there is no transition owner, the effort is not ready for insertion. It may be ready for exploration. It may be ready for research. It is not ready for an adoption push.

This distinction protects everyone. It protects operators from being handed tools no one can sustain. It protects program offices from absorbing unmanaged risk. It protects vendors from confusing access with transition. It protects executives from mistaking activity for progress.

Transition ownership is not an administrative detail. It is the bridge between possibility and capability.

Governed AI Raises the Bar.

AI and autonomy make this management problem sharper.

An AI-enabled insertion is not only a question of performance. It is a question of authority, evidence, monitoring, and change control.

The organization has to know what the system is allowed to recommend, decide, coordinate, or execute. It has to know when human judgment is required, when human review is meaningful, and when the speed or complexity of the system makes review performative. It has to know what data conditions invalidate the output, what drift looks like, what failure modes matter, and who can suspend, roll back, or narrow the capability after deployment.

The NIST AI Risk Management Framework describes AI risk management as a lifecycle activity across govern, map, measure, and manage functions [4]. That is the right posture for insertion. Risk is not retired at demo completion. It changes as the system meets new users, new data, new adversary behavior, and new operational constraints.

For autonomy, DoD Directive 3000.09 reinforces the importance of verification, validation, testing, human judgment, and governance in high-consequence autonomy contexts [5]. Even when the technology is not a weapon system, the management lesson holds. Authority cannot be an afterthought.

A system that recommends, acts, coordinates, or delegates needs boundaries before it scales.

Evidence Has to Feed the Next Decision.

Technology insertion should be a closed loop.

The organization senses a mission need, frames the problem, tests fit, defines acceptance criteria, runs the pilot, evaluates evidence, decides scale or stop, and feeds the result back into the next

insertion decision.

That loop is what separates a management system from demo theater.

If the pilot works, the organization should leave with more than a success story. It should know which architecture choices mattered, which data assumptions held, which users trusted the system, which authorities were required, which contract path worked, and what evidence made scale credible.

If the pilot fails, the organization should still get value. It should know whether the problem was weak ownership, poor fit, immature technology, unrealistic acceptance criteria, missing funding, cyber friction, or a transition path that never existed.

Learning is the product even when adoption is not.

Scale or Stop.

The most neglected insertion decision is the stop decision.

Stopping is not cynicism. It is discipline.

A pilot should stop when the mission owner is unclear, the integration path is unrealistic, the evidence does not support confidence, the acquisition path is absent, the sustainment model is fantasy, or the technology solves a problem that is not important enough.

Scale should be just as explicit. Scale means the organization has evidence, ownership, architecture, funding, training, cyber posture, monitoring, and a reusable pattern. It does not mean the demo looked good.

Technology insertion is not a demo day.

It is a management system for turning technical possibility into operational capability.

If the system is missing, the demo becomes theater.

If the system works, the organization learns faster, transitions more honestly, and stops confusing access to technology with adoption of capability.

Sources

1. U.S. Department of Defense, DoD Instruction 5000.87, Operation of the Software Acquisition Pathway.
2. U.S. Department of Defense, DoD Instruction 5000.97, Digital Engineering.
3. U.S. Government Accountability Office, Technology Readiness Assessment Guide, GAO-20-48G.
4. National Institute of Standards and Technology, AI Risk Management Framework.
5. U.S. Department of Defense, DoD Directive 3000.09, Autonomy in Weapon Systems.